

GEOGRAPHY

INDIAN GEOGRAPHY



INDIAN SOILS

MODULE - 9

**SOIL
CONSERVATION**

SOIL CONSERVATION



- 1** Soil conservation includes all those measures which help in protecting the soil from **erosion and exhaustion**.
- 2** Soil erosion has been continuing over such a large part of India for such a long time that it **has assumed alarming proportions**.
- 3** Unfortunately, **it has not attracted the attention that it deserves**. Our peasantry is not fully aware of many benefits of soil conservation and neglect of soil is like killing the hen that lays the golden eggs. Following methods are normally adopted for conserving soil.

METHODS ADOPTED FOR SOIL CONSERVATION

METHOD 1

AFFORESTATION



1 The best way to conserve soil is to **increase area under forests**. Indiscriminate felling of trees should be stopped and efforts should be made to plant trees in new areas.

2 A minimum area of forest land for the whole country that is considered healthy for soil and water conservation is **between 20 to 25%** but it was raised to **33%** in the **second five year plan**; the proportion being 20 % for the plains and 60% for hilly and mountainous region.

METHODS ADOPTED FOR SOIL CONSERVATION

CHECKING OVERGRAZING

METHOD 2



1 Overgrazing of forests and grass lands by animals, especially by goats and sheep, should be properly checked.

2 Separate grazing grounds should be **earmarked** and fodder crops should be grown in larger quantities.

3 Animals freely move about in the fields for grazing and spoil the soil by their hoofs which leads to soil erosion. This should be avoided.

METHODS ADOPTED FOR SOIL CONSERVATION

CONSTRUCTING DAMS

METHOD 3



1 Much of soil erosion by river floods can be avoided by constructing dams across the rivers.

2 This checks the speed of water and saves soil from erosion.

METHODS ADOPTED FOR SOIL CONSERVATION

METHOD 4

CHANGING AGRICULTURAL PRACTICES

We can save lot of our valuable soil by bringing about certain changes in our agricultural practices.

1 CROP ROTATION



- 1 In many parts of India, a particular crop is sown in the same field year after year. This practice takes away certain elements from the soil, making it infertile and exhausted rendering it unsuitable for that crop.
- 2 Rotation of crops is the system in which a different crop is cultivated on a piece of land each year.
- 3 This helps to conserve soil fertility as different crops make different demands in the soil.

METHODS ADOPTED FOR SOIL CONSERVATION

CHANGING AGRICULTURAL PRACTICES

METHOD 4

1 CROP ROTATION



4 For example, potatoes require much potash but wheat requires nitrate. Thus, it is best to alternate crops in the field.

5 **Legumes** such as peas, beans, clover, vetch and many other plants, add **nitrites** to the soil by converting free nitrogen in the air into nitrogenous nodules on their roots.

6 Thus, if they are included in the crop rotation nitrogenous fertilizers can be dispensed with.

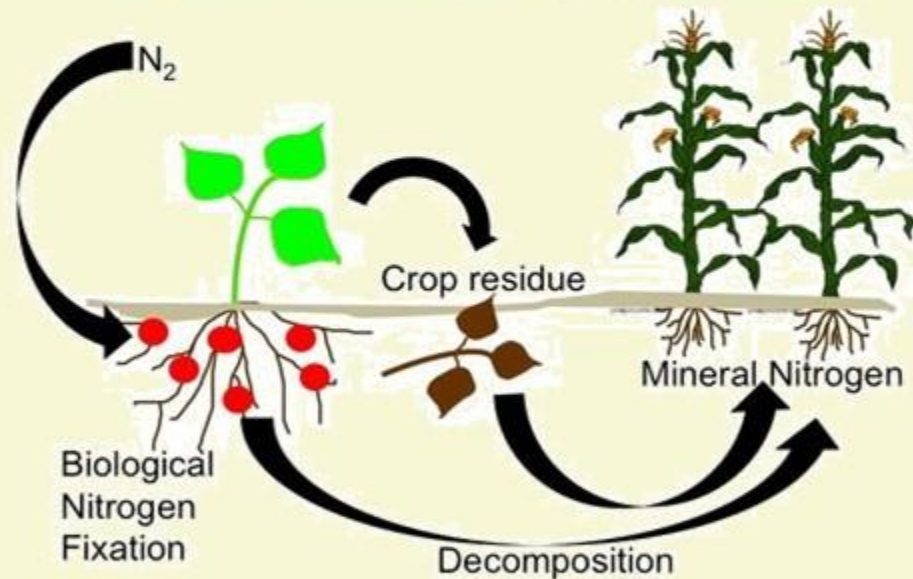
METHODS ADOPTED FOR SOIL CONSERVATION

CHANGING AGRICULTURAL PRACTICES

METHOD 4

1 CROP ROTATION

Legume based cropping system



7 By rotating different types of crops in successive years, soil fertility can be naturally maintained.

8 For example, wheat may be cultivated in the first year, barley in the second and legumes in the third.

9 The cycle may be then repeated. Further, there are some crops such as **maize, cotton, tobacco and potato** which can be classified as erosion inducing, whilst some other crops such as grass, forage crops and many legumes are erosion resisting.

10 Small grain crops like wheat, barley, oats and rice are between these two examples.

METHODS ADOPTED FOR SOIL CONSERVATION

CHANGING AGRICULTURAL PRACTICES

METHOD 4

2 STRIP CROPPING

- 1 Crops may be cultivated in latent strips, parallel to one another.



- 2 Some strips may be allowed to lie fallow while in others different crops may be sown e.g. grains, legumes, small tree crops, grass etc.
- 3 Various crops ripen at different times of the year and are harvested at intervals.
- 4 This ensures that at no time of the year the entire area is left bare or exposed.
- 5 The tall growing crops act as wind breaks and the strips which are often parallel to the contours help in increasing water absorption by the soil by slowing down run off.

METHODS ADOPTED FOR SOIL CONSERVATION

CHANGING AGRICULTURAL PRACTICES

METHOD 4

3 USE OF EARLY MATURING VARIETIES



Early maturing varieties of crops take less time to mature and thus put lesser pressure on the soil. In this way it can help in reducing the soil erosion.

METHODS ADOPTED FOR SOIL CONSERVATION

CHANGING AGRICULTURAL PRACTICES

METHOD 4

4 CONTOUR PLOUGHING



1 If ploughing is done at right angles to the hill slope, following the natural contours of the hill, the ridges and furrows break the flow of water down the hill.

2 This prevents excessive soil loss as gullies are less likely to develop and also reduce run-off so that plants receive more water.

3 Thus, by growing crops in contour pattern, plants can absorb much of the rain water and erosion is minimized. When viewed from above, the field looks like a contour map.

METHODS ADOPTED FOR SOIL CONSERVATION

METHOD 4

CHANGING AGRICULTURAL PRACTICES

5 TERRACING AND CONTOUR BUNDING



Terrace farming or Terracing

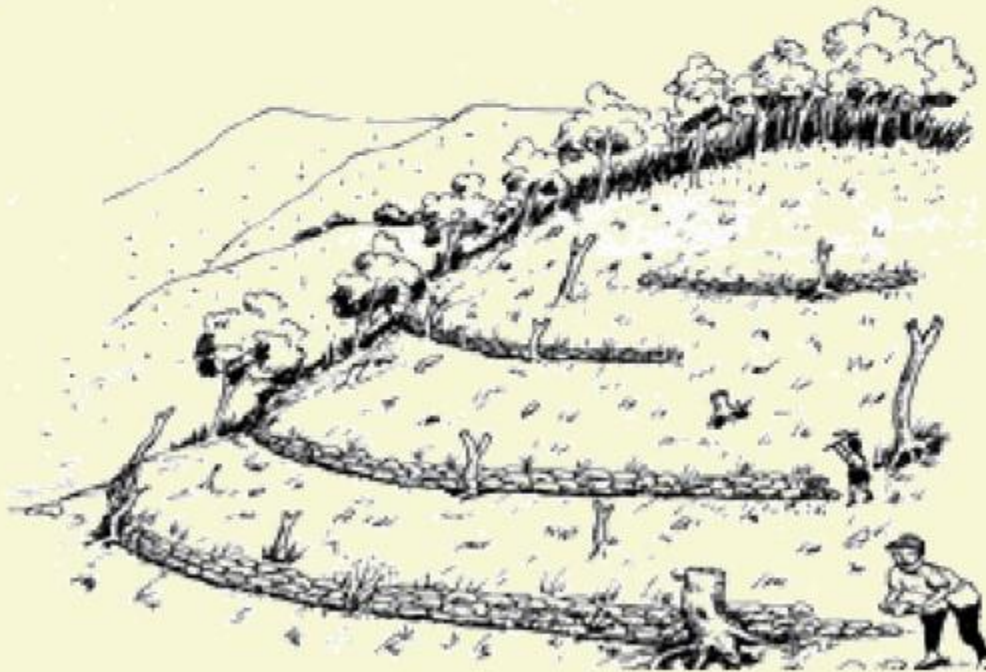
- 1 Terracing and Contour Bunding across the hill slopes is a very effective and one of the oldest methods of soil conservation.
- 2 Hill slope is cut into a number of terraces having horizontal top and steep slopes on the back and front.
- 3 Contour Bunding involves the construction of banks/bunds along the contours.

METHODS ADOPTED FOR SOIL CONSERVATION

CHANGING AGRICULTURAL PRACTICES

METHOD 4

5 TERRACING AND CONTOUR BUNDING



Contour bunding

4 Terracing and Contour Bunding which divides the hill slope into numerous small slopes, checks the flow water, promotes absorption of water by soil and saves soil from erosion.

5 Sometimes tree crops such as rubber are also planted to combat soil erosion.

6 But, there is a limit to which bunding is an effective measure of soil conservation.

METHODS ADOPTED FOR SOIL CONSERVATION

METHOD 4

CHANGING AGRICULTURAL PRACTICES

5 TERRACING AND CONTOUR BUNDING



Contour bunding

7 When the slope is steeper than 8% or 1 in 12, bunding becomes expensive and less effective.

8 Steep slopes of over 20% or 1 in 5 should not be terraced at all. Fields of a slope steeper than 15 % or 1 in 6 should be withdrawn from ploughing as they are not usually worth the labour of making benches very close together.

METHODS ADOPTED FOR SOIL CONSERVATION

METHOD 4

CHANGING AGRICULTURAL PRACTICES

6 CHECKING SHIFTING CULTIVATION



1 Checking and reducing shifting cultivation by persuading the tribal people to switch over to settled agriculture is a very effective method of soil conservation.

2 This can be done by making arrangements for their resettlement which involves the provision of residential, accommodation agricultural implements, seeds, manures, cattle and reclaimed land.

METHODS ADOPTED FOR SOIL CONSERVATION

CHANGING AGRICULTURAL PRACTICES

METHOD 4

7 PLOUGHING THE LAND IN RIGHT DIRECTION



Ploughing to be done across the slope

Ploughing the land in a direction also protects the top soil from erosion.

INDIAN GEOGRAPHY

PLAN OF ACTION



1. INDIAN MONSOON
2. INDIAN SOILS
3. INDIAN NATURAL VEGETATION
4. INDIAN PHYSIOGRAPHY
 - HIMALAYAN RANGES
 - NORTHERN PLAINS
 - PENINSULAR PLATEAU
 - COASTAL PLAINS OF INDIA
 - ISLANDS OF INDIA
 - HILL RANGES OF THE PENINSULAR PLATEAU



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